

DBMS practical Test

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Database Creation

```
CREATE DATABASE DBMS_Practical;
```

Select the Database

```
USE DBMS_Practical;
```

Create COURSE Table

```
CREATE TABLE COURSE (  
    Course_ID INT PRIMARY KEY,  
    Course_Name VARCHAR(50),  
    Fees DECIMAL(10,2)  
);
```

Create STUDENT Table

```
CREATE TABLE STUDENT (  
    Student_ID INT PRIMARY KEY,  
    Student_Name VARCHAR(50),  
    Course_ID INT,  
    Marks INT  
);
```

INSERT

```
INSERT INTO COURSE VALUES
```

```
(101, 'Data Science', 50000),
```

```
(102, 'Computer Science', 45000),
```

```
(103, 'Information Technology', 40000),
```

```
(104, 'Artificial Intelligence', 55000);
```

```
SELECT * FROM COURSE;
```

```
INSERT INTO STUDENT VALUES
```

```
(1, 'Aarav', 101, 85),
```

```
(2, 'Sneha', 102, 72),
```

```
(3, 'Rahul', 101, 65),
```

```
(4, 'Priya', 103, 90),
```

```
(5, 'Rohan', 102, 60),
```

```
(6, 'Ananya', 104, 78),
```

```
(7, 'Neha', 101, 75),
```

```
(8, 'Aditya', 103, 55);
```

```
SELECT * FROM STUDENT;
```

Q1)b)Display students who have scored more than 70 marks.

```
SELECT *
```

```
FROM STUDENT
```

```
WHERE Marks > 70;
```

Student_ID	Student_Name	Course_ID	Marks
1	Aarav	101	85
2	Sneha	102	72
4	Priya	103	90
6	Ananya	104	78
7	Neha	101	75

Q1(c): Display student names with their course names

```
SELECT s.Student_Name, c.Course_Name
```

FROM STUDENT s

JOIN COURSE c

ON s.Course_ID = c.Course_ID;

The screenshot shows the phpMyAdmin interface with the 'STUDENT' table selected. A SQL query is entered in the 'Show query box' and executed. The query is: `SELECT s.Student_Name, c.Course_Name FROM STUDENT s JOIN COURSE c ON s.Course_ID = c.Course_ID;`. The result shows 8 rows of data. Below the table, there are controls for 'Number of rows' (set to 25), 'Filter rows' (Search this table), and 'Sort by key' (None).

Student_Name	Course_Name
Aarav	Data Science
Sneha	Computer Science
Rahul	Data Science
Priya	Information Technology
Rohan	Computer Science
Ananya	Artificial Intelligence
Neha	Data Science
Aditya	Information Technology

Q1(d): Find average marks for each Course_ID

SELECT Course_ID, AVG(Marks) AS Average_Marks

FROM STUDENT

GROUP BY Course_ID;

The screenshot shows the phpMyAdmin interface with the 'STUDENT' table selected. A SQL query is entered in the 'Show query box' and executed. The query is: `SELECT Course_ID, AVG(Marks) AS Average_Marks FROM STUDENT GROUP BY Course_ID;`. The result shows 4 rows of data. Below the table, there are controls for 'Number of rows' (set to 25), 'Filter rows' (Search this table), and 'Sort by key' (None).

Course_ID	Average_Marks
101	75.0000
102	66.0000
103	72.5000
104	78.0000

Q1)E)Display only those courses whose average marks are greater than 65 using HAVING.

SELECT Course_ID, AVG(Marks) AS Average_Marks

FROM STUDENT

GROUP BY Course_ID

HAVING AVG(Marks) > 65;

The screenshot shows the phpMyAdmin interface. The left sidebar lists databases including 'dbms_practical'. The main panel displays a SQL query: `SELECT Course_ID, AVG(Marks) AS Average_Marks FROM STUDENT GROUP BY Course_ID HAVING AVG(Marks) > 65;`. Below the query, the results are shown in a table with two columns: 'Course_ID' and 'Average_Marks'. The results table contains four rows of data.

Course_ID	Average_Marks
101	75.0000
102	66.0000
103	72.5000
104	78.0000

Q2: Create EMPLOYEE table

CREATE TABLE EMPLOYEE (

Emp_ID INT PRIMARY KEY,

Emp_Name VARCHAR(50),

Department VARCHAR(50),

Salary DECIMAL(10,2),

Joining_Date DATE

);

DESC EMPLOYEE;

INSERT INTO EMPLOYEE VALUES

(1, 'Amit', 'IT', 60000, '2024-01-15'),

(2, 'Neha', 'HR', 45000, '2023-06-20'),

(3, 'Rahul', 'IT', 75000, '2022-03-10'),

(4, 'Priya', 'Finance', 55000, '2024-02-05'),

(5, 'Sneha', 'IT', 65000, '2023-08-12'),

(6, 'Rohan', 'HR', 40000, '2025-01-18'),

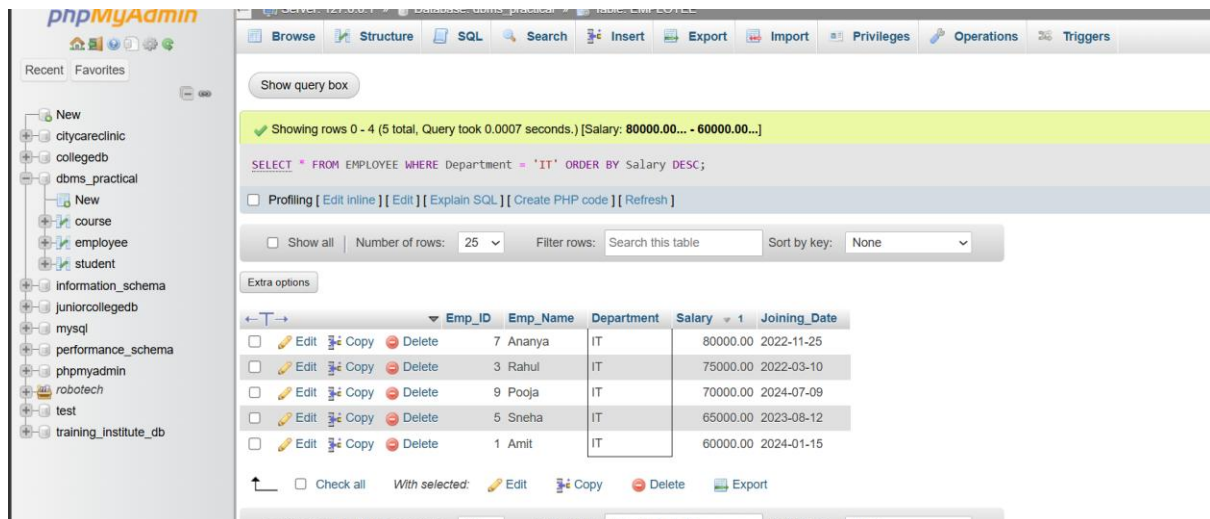
(7, 'Ananya', 'IT', 80000, '2022-11-25'),

(8, 'Karan', 'Finance', 60000, '2023-04-14'),

```
(9, 'Pooja', 'IT', 70000, '2024-07-09'),  
(10, 'Vikas', 'Sales', 50000, '2023-09-30');  
  
SELECT * FROM EMPLOYEE;
```

Q2)A)Display employees belonging to the IT department in descending order of salary.

```
SELECT *  
  
FROM EMPLOYEE  
  
WHERE Department = 'IT'  
  
ORDER BY Salary DESC;
```



Showing rows 0 - 4 (5 total, Query took 0.0007 seconds.) [Salary: 80000.00... - 60000.00...]

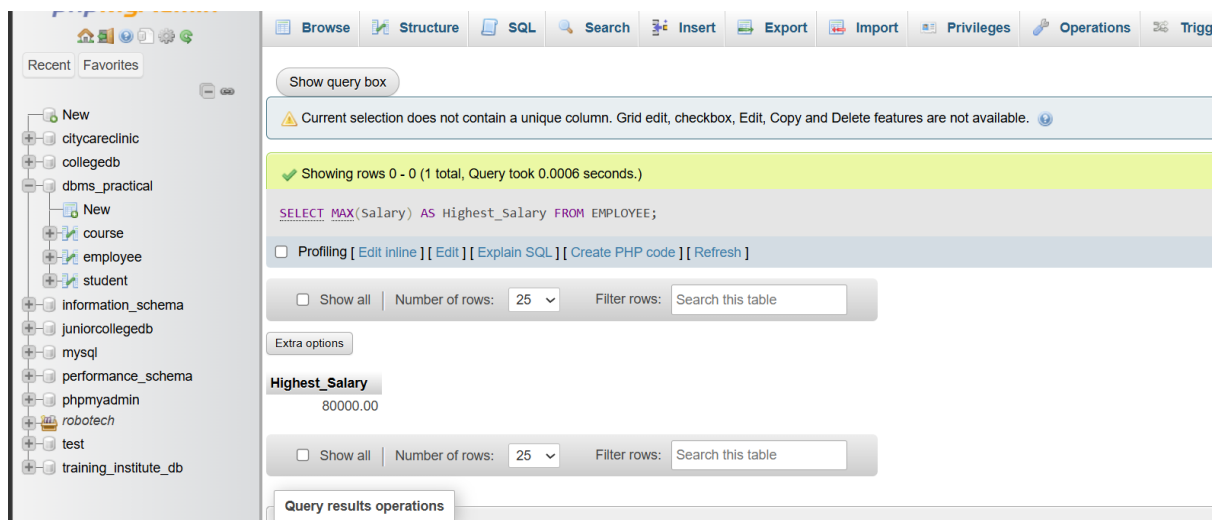
```
SELECT * FROM EMPLOYEE WHERE Department = 'IT' ORDER BY Salary DESC;
```

Number of rows: 25 Filter rows: Search this table Sort by key: None

	Emp_ID	Emp_Name	Department	Salary	Joining_Date
☐	7	Ananya	IT	80000.00	2022-11-25
☐	3	Rahul	IT	75000.00	2022-03-10
☐	9	Pooja	IT	70000.00	2024-07-09
☐	5	Sneha	IT	65000.00	2023-08-12
☐	1	Amit	IT	60000.00	2024-01-15

Q2(b): Find the highest salary

```
SELECT MAX(Salary) AS Highest_Salary  
  
FROM EMPLOYEE;
```



Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 0 (1 total, Query took 0.0006 seconds.)

```
SELECT MAX(Salary) AS Highest_Salary FROM EMPLOYEE;
```

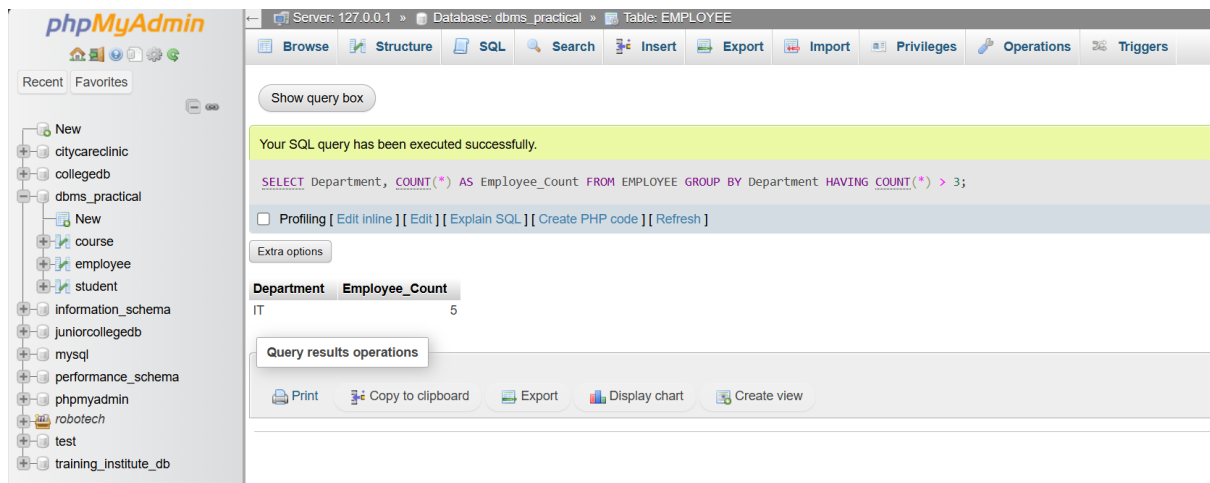
Number of rows: 25 Filter rows: Search this table

Highest_Salary
80000.00

Query results operations

Q2(c): Departments having more than 3 employees

```
SELECT Department, COUNT(*) AS Employee_Count  
FROM EMPLOYEE  
GROUP BY Department  
HAVING COUNT(*) > 3;
```



The screenshot shows the phpMyAdmin interface. The left sidebar displays a database structure with 'dbms_practical' selected, containing tables like 'course', 'employee', and 'student'. The main panel shows the 'Table: EMPLOYEE' view. A message states 'Your SQL query has been executed successfully.' Below this, the executed query is displayed: `SELECT Department, COUNT(*) AS Employee_Count FROM EMPLOYEE GROUP BY Department HAVING COUNT(*) > 3;`. A table of results is shown with two columns: 'Department' and 'Employee_Count'. The first row shows 'IT' with a count of 5. At the bottom, there are buttons for 'Print', 'Copy to clipboard', 'Export', 'Display chart', and 'Create view'.

Department	Employee_Count
IT	5

Q2(d): Employees earning more than the average salary

```
SELECT *  
FROM EMPLOYEE  
WHERE Salary > (  
    SELECT AVG(Salary)  
    FROM EMPLOYEE  
);
```

phpMyAdmin

Server: 127.0.0.1 » Database: dbms_practical » Table: EMPLOYEE

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Show query box

Showing rows 0 - 3 (4 total, Query took 0.0008 seconds.)

```
SELECT * FROM EMPLOYEE WHERE Salary > ( SELECT AVG(Salary) FROM EMPLOYEE );
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	Emp_ID	Emp_Name	Department	Salary	Joining_Date
☐ Edit Copy Delete	3	Rahul	IT	75000.00	2022-03-10
☐ Edit Copy Delete	5	Sneha	IT	65000.00	2023-08-12
☐ Edit Copy Delete	7	Ananya	IT	80000.00	2022-11-25
☐ Edit Copy Delete	9	Pooja	IT	70000.00	2024-07-09

☐ Check all | With selected: Edit Copy Delete Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Q2(e): Create a VIEW

CREATE VIEW IT_Employees AS

SELECT *

FROM EMPLOYEE

WHERE Department = 'IT';

phpMyAdmin

Server: 127.0.0.1 » Database: dbms_practical » Table: EMPLOYEE

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Show query box

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0095 seconds.)

```
CREATE VIEW IT_Employees AS SELECT * FROM EMPLOYEE WHERE Department = 'IT';
```

[Edit inline] [Edit] [Create PHP code]

phpMyAdmin

Server: 127.0.0.1 » Database: dbms_practical » View: IT_Employees

Browse Structure SQL Search Insert Export Privileges Operations

Show query box

⚠ Current selection does not contain a unique column. Grid edit, Edit, Copy and Delete features may result in undesired behavior.

✓ Showing rows 0 - 4 (5 total, Query took 0.0004 seconds.)

`SELECT * FROM IT_Employees;`

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

	Emp_ID	Emp_Name	Department	Salary	Joining_Date
☐ Edit Copy Delete	1	Amit	IT	60000.00	2024-01-15
☐ Edit Copy Delete	3	Rahul	IT	75000.00	2022-03-10
☐ Edit Copy Delete	5	Sneha	IT	65000.00	2023-08-12
☐ Edit Copy Delete	7	Ananya	IT	80000.00	2022-11-25
☐ Edit Copy Delete	9	Pooja	IT	70000.00	2024-07-09

☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Console

phpMyAdmin

Server: 127.0.0.1 » Database: dbms_practical » View: IT_Employees

Browse Structure SQL Search Insert Export Privileges Operations

Show query box

⚠ Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

✓ Your SQL query has been executed successfully.

`SHOW FULL TABLES WHERE TABLE_TYPE = 'VIEW';`

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

Extra options

Tables_in_dbms_practical	Table_type
it_employees	VIEW

Query results operations

[Print](#) [Copy to clipboard](#) [Create view](#)

Final summary of the questions

-- Q1(a)

CREATE TABLE STUDENT (

Student_ID INT PRIMARY KEY,

Student_Name VARCHAR(50),

Course_ID INT,

Marks INT

);

-- Q1(b)

```
SELECT *  
FROM STUDENT  
WHERE Marks > 70;
```

-- Q1(c)

```
SELECT s.Student_Name, c.Course_Name  
FROM STUDENT s  
JOIN COURSE c  
ON s.Course_ID = c.Course_ID;
```

-- Q1(d)

```
SELECT Course_ID, AVG(Marks) AS Average_Marks  
FROM STUDENT  
GROUP BY Course_ID;
```

-- Q1(e)

```
SELECT Course_ID, AVG(Marks) AS Average_Marks  
FROM STUDENT  
GROUP BY Course_ID  
HAVING AVG(Marks) > 65;
```

Q2 — EMPLOYEE

-- Q2(a)

```
SELECT *  
FROM EMPLOYEE  
WHERE Department = 'IT'
```

```
ORDER BY Salary DESC;
```

```
-- Q2(b)
```

```
SELECT MAX(Salary) AS Highest_Salary  
FROM EMPLOYEE;
```

```
-- Q2(c)
```

```
SELECT Department, COUNT(*) AS Employee_Count  
FROM EMPLOYEE  
GROUP BY Department  
HAVING COUNT(*) > 3;
```

```
-- Q2(d)
```

```
SELECT *  
FROM EMPLOYEE  
WHERE Salary > (  
    SELECT AVG(Salary)  
    FROM EMPLOYEE  
);
```

```
-- Q2(e)
```

```
CREATE VIEW IT_Employees AS  
SELECT *  
FROM EMPLOYEE  
WHERE Department = 'IT';
```

